

# ORDER-DEPENDENT NONLINEAR CLOSED-ORBIT RESPONSE IN A CESR MODEL

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## Abstract

**DRAFT PLACEHOLDER.** Write after the remaining sextupole/octupole attribution scan is complete and the result set is frozen.

## INTRODUCTION

**DRAFT PLACEHOLDER.** Add the motivation, the relation to linear orbit-response matrices and nonlinear closed-orbit methods, and a concise statement of the paper's contribution after the result set is frozen.

## ORBIT AND OPTICS CALCULATION

A Bmad-compatible model of the Cornell Electron Storage Ring (CESR) was used with 119 corrector variables: 58 horizontal and 61 vertical correctors. The orbit benchmark contains 1000 deterministic input samples, including the zero-control reference. The nonzero inputs are independently sampled Gaussian corrector changes with an rms value of  $5\text{ }\mu\text{rad}$ , clipped at three standard deviations. Closed orbits are recorded at 99 zero-length detector markers in both transverse planes.

SciBmad solves the radio-frequency-on closed orbits as a batch. A cached  $6 \times 119$  nominal closed-orbit response matrix supplies the initial estimate  $z_0 + R\Delta k$ ; Newton iterations reuse the nominal Jacobian and send failed lanes to a full automatic-differentiation fallback. All 1000 formal samples converged without fallback. On the common `lnx201` host, the maintained SciBmad run required 33.178 s, compared with 67.370 s for Bmad/Tao. The 198000 compared detector coordinates had correlation 0.999999966 and a maximum absolute difference of  $8.14\text{ }\mu\text{m}$ . The latter includes differences between the independently represented lattices and is not a Newton closure tolerance.

The associated radio-frequency-off optics benchmark evaluates periodic coupled Twiss quantities and their first momentum derivatives at the same 99 markers. Table 1 summarizes representative formal results. The parameterized calculation uses one third-order phase-space map with 119 first-order corrector parameters; it is a local surrogate rather than 1000 independent optics solutions.

## ORBIT RESPONSE ERROR ANALYSIS

For corrector perturbation  $\Delta k$ , the exact detector orbit is expanded about the nominal machine as

$$u(\Delta k) = u_0 + J\Delta k + \frac{1}{2}H[\Delta k, \Delta k] + \frac{1}{6}T[\Delta k, \Delta k, \Delta k] + \dots \quad (1)$$

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Table 1: Representative 1000-Sample Calculation Results

| Calculation          | Time [s] | Rate [s <sup>-1</sup> ] | Median corr. |
|----------------------|----------|-------------------------|--------------|
| Bmad RF-on orbit     | 67.370   | 14.843                  | 1.000000     |
| SciBmad RF-on orbit  | 33.178   | 30.140                  | 1.000000     |
| SciBmad point optics | 570.771  | 1.752                   | 1.000000     |
| Parameterized optics | 73.130   | 13.674                  | 0.999996     |

The residual of the nominal orbit-response approximation is  $e = u(\Delta k) - u_0 - J\Delta k$ . Inputs are written as  $\Delta k = \rho d$  ( $5\text{ }\mu\text{rad}$ ), where every active part of  $d$  has unit rms. A sweep over 600 fixed Gaussian directions compares all-corrector, horizontal-only, and vertical-only inputs. It contains 36001 nonlinear states: one reference plus 20 radii, 600 directions, and three input scenarios. Every state converged through  $\rho = 12.8$ .

The mean detector root-mean-square error  $E(\rho)$  is calculated separately for the horizontal and vertical detector coordinates. Quadratic response is diagnosed by constant  $E/\rho^2$  and a local log-log slope near two. The small-radius coefficients in Table 2 show two strong anisotropies. First, the all-corrector horizontal residual is close to the vertical-only value. Second, the all-corrector vertical coefficient is about 73 times the vertical-only coefficient and 145 times the horizontal-only coefficient. The latter cannot be reproduced by either isolated plane and therefore motivates a direct mixed-Hessian test.

Table 2: Mean Quadratic Coefficients at  $\rho = 0.1$

| Active correctors | $x$ residual<br>[ $\mu\text{m}/\rho^2$ ] | $y$ residual<br>[ $\mu\text{m}/\rho^2$ ] |
|-------------------|------------------------------------------|------------------------------------------|
| All               | 0.5648                                   | 0.5321                                   |
| Horizontal only   | 0.1570                                   | 0.003660                                 |
| Vertical only     | 0.5524                                   | 0.007270                                 |

The all-corrector channels remain near quadratic through the complete common range. Horizontal-only remains near quadratic through  $\rho = 64$ . The clear exception is the vertical detector residual for vertical-only inputs: it leaves the quadratic guide around  $\rho = 0.28\text{--}0.4$ , and its slope approaches three while the full nonlinear reference remains converged.

## ALL-CORRECTOR SECOND-ORDER SOURCE

The mixed block was isolated with 100 fixed pairs of independently sampled horizontal and vertical corrector directions. Each direction was normalized to unit rms in its active family and reused at eight radii over  $0.1 \leq \rho \leq 1.13$ .

Pure positive and negative states and all four joint signs were evaluated, giving 6401 converged nonlinear states without fallback. For the full detector vector,

$$Q_{hh} = \frac{u(+h, 0) + u(-h, 0)}{2} - u_0, \quad (2)$$

$$Q_{vv} = \frac{u(0, +v) + u(0, -v)}{2} - u_0, \quad (3)$$

$$Q_{hv} = \frac{u(+h, +v) - u(+h, -v) - u(-h, +v) + u(-h, -v)}{4}. \quad (4)$$

Each signed quadratic residual was reconstructed as  $Q_{hh} + Q_{vv} + s_h s_v Q_{hv}$  before taking detector rms norms. For orbit component  $u \in \{x, y\}$ , the tabulated block shares are calculated for each direction before taking percentiles,

$$f_{ab,u} = \frac{\|Q_{ab,u}\|^2}{\|Q_{hh,u}\|^2 + \|Q_{hv,u}\|^2 + \|Q_{vv,u}\|^2}, \quad (5)$$

$ab \in \{hh, hv, vv\}.$

The vertical  $Q_{hv}/\rho^2$  coefficient changes by only 0.000423% across the fit interval. At  $\rho = 1.13$ , the mean vertical rms components are 0.004 72  $\mu\text{m}$ , 0.009 83  $\mu\text{m}$ , and 0.714 75  $\mu\text{m}$  for  $Q_{hh}$ ,  $Q_{vv}$ , and  $Q_{hv}$ , respectively. Table 3 shows the distinct physical result for the two orbit components. For  $x$ , the  $vv$  share has 10th, 50th, and 90th percentiles of 61.72%, 90.83%, and 98.15%. For  $y$ , the corresponding  $hv$  percentiles are 99.930%, 99.977%, and 99.991%. The signed reconstruction leaves 0.953% of the exact vertical residual rms, while adding  $Q_{hv}$  reduces the mean squared reconstruction error by 99.982% relative to the pure-block reconstruction. By contrast, the horizontal mixed share is only 0.0146%. Thus the all-corrector vertical quadratic excess is a mixed horizontal-vertical response, rather than the sum of the isolated pure-plane terms.

## VERTICAL-CORRECTOR SIGNED PARITY

For a midplane-symmetric ideal lattice, reversal of the vertical-corrector vector gives

$$x(v) = x(-v), \quad y(v) = -y(-v). \quad (6)$$

Consequently, the vertical detector response to vertical-only input contains odd powers of  $v$ . After subtraction of the linear response, its leading allowed nonlinear term is cubic. By contrast, the horizontal detector response permits a quadratic  $v^2$  term. This is the closed-orbit form of the normal-multipole midplane selection rule [1].

The order was tested directly with 100 fixed 61-dimensional vertical-corrector directions and 27 radii. Both signs of every direction were evaluated, giving 5401 converged nonlinear states with no fallback and a maximum one-turn closure norm of  $9.994 \times 10^{-14}$ . For detector plane  $u$ , the paired components are

$$e_{\text{even}} = \frac{u(+\Delta k) + u(-\Delta k)}{2} - u_0, \quad (7)$$

$$e_{\text{odd, nl}} = \frac{u(+\Delta k) - u(-\Delta k)}{2} - J\Delta k. \quad (8)$$

Symmetric positive and negative corrector excitations are also the standard central-difference construction used for measured orbit-response matrices [2].

The vertical even component follows  $\rho^2$ , whereas the odd nonlinear component follows  $\rho^3$ . Their mean detector rms values cross at  $\rho \approx 1.31$ , corresponding to an active-corrector rms kick of about 6.5  $\mu\text{rad}$ . At  $\rho = 6.4$ , the mean odd/even ratio is 4.91. The horizontal detector control remains overwhelmingly even: its odd/even ratio is  $3.05 \times 10^{-4}$  at the same radius.

The transition is direction dependent. At  $\rho = 6.4$ , the median odd/even ratio is 2.48 and 80% of directions are odd dominated. At the extension point  $\rho = 10.05$ , the 10th, 50th, and 90th percentiles of  $f$  are 50.06%, 93.84%, and 99.76%, respectively. Thus at least 90% of the sampled directions are odd dominated at that amplitude.

## Multipole attribution

**DRAFT PLACEHOLDER.** Insert the normal-sextupole and octupole ablation/strength scans. Recompute the linear response for every modified lattice and fit the direction-resolved cubic response vector, rather than comparing only one large-amplitude rms value. Test a sextupole-cascade scaling with  $K_2^2$  against a direct octupole scaling with  $K_3$ , including interference if both are significant.

## IMPLICATIONS FOR RESPONSE SURROGATES

Equation (1) shows why one global truncation order is not sufficient. A quadratic Hessian is appropriate for channels whose residuals retain  $\rho^2$  scaling over the intended input distribution. It cannot represent the vertical-only vertical response after the odd cubic component becomes comparable to the symmetry-breaking even term. A practical surrogate should therefore select order by input/output channel, retain exact nonlinear solutions as validation references, and report its validity radius together with direction-dependent error statistics. Nonlinear closed-orbit and higher-order response measurements have similarly been used to identify sextupole behavior and other nonlinear lattice information [3–5].

## FUTURE OPTICS ERROR ANALYSIS

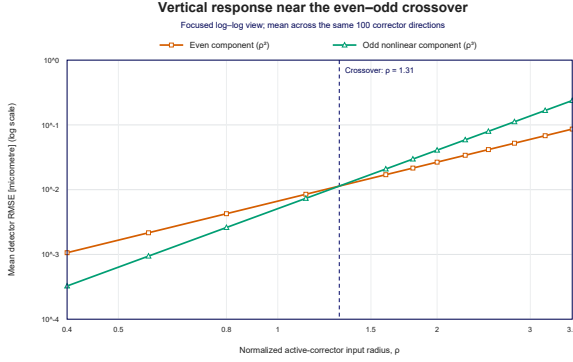
The same order-versus-amplitude analysis will be applied to ordinary optics, chromatic quantities, coupling observables, and orbit-derived fields. These classes will be reported separately because aggregation can hide a small quadratic coefficient or an early higher-order transition. This extension is future work and is not used to support the orbit conclusions above.

## CONCLUSION

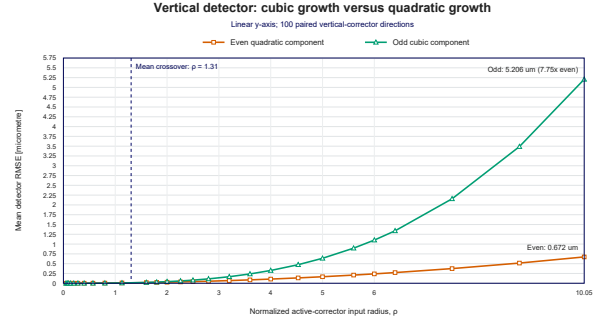
The CESR-model benchmarks establish reproducible closed-orbit and optics calculations on a shared 1000-sample

Table 3: Direction-resolved quadratic-block squared-norm shares. Each entry is median [P10, P90] in percent across the same 100 fixed H/V direction pairs. The values are invariant at the displayed precision over the verified quadratic interval.

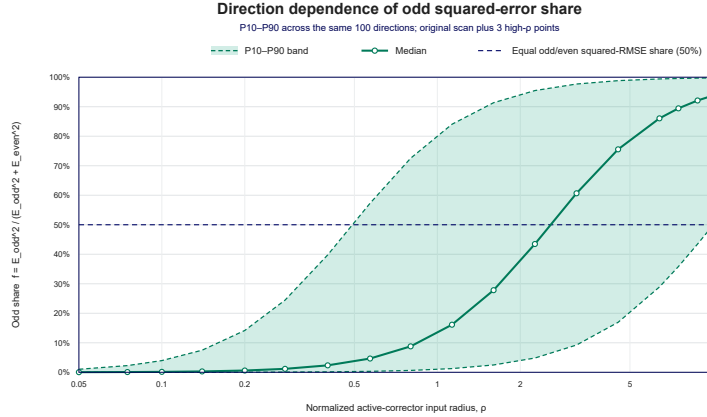
| $\rho$ | X orbit component  |                         |                      | Y orbit component       |                         |                         |
|--------|--------------------|-------------------------|----------------------|-------------------------|-------------------------|-------------------------|
|        | $hh$               | $hv$                    | $vv$                 | $hh$                    | $hv$                    | $vv$                    |
| 0.1    | 9.14 [1.83, 38.26] | 0.0088 [0.0017, 0.0313] | 90.83 [61.72, 98.15] | 0.0040 [0.0007, 0.0231] | 99.977 [99.930, 99.991] | 0.0163 [0.0044, 0.0651] |
| 0.14   | 9.14 [1.83, 38.26] | 0.0088 [0.0017, 0.0313] | 90.83 [61.72, 98.15] | 0.0040 [0.0007, 0.0231] | 99.977 [99.930, 99.991] | 0.0163 [0.0044, 0.0651] |
| 0.2    | 9.14 [1.83, 38.26] | 0.0088 [0.0017, 0.0313] | 90.83 [61.72, 98.15] | 0.0040 [0.0007, 0.0231] | 99.977 [99.930, 99.991] | 0.0163 [0.0044, 0.0651] |
| 0.28   | 9.14 [1.83, 38.26] | 0.0088 [0.0017, 0.0313] | 90.83 [61.72, 98.15] | 0.0040 [0.0007, 0.0231] | 99.977 [99.930, 99.991] | 0.0163 [0.0044, 0.0651] |
| 0.4    | 9.14 [1.83, 38.26] | 0.0088 [0.0017, 0.0313] | 90.83 [61.72, 98.15] | 0.0040 [0.0007, 0.0231] | 99.977 [99.930, 99.991] | 0.0163 [0.0044, 0.0651] |
| 0.57   | 9.14 [1.83, 38.26] | 0.0088 [0.0017, 0.0313] | 90.83 [61.72, 98.15] | 0.0040 [0.0007, 0.0231] | 99.977 [99.930, 99.991] | 0.0163 [0.0044, 0.0651] |
| 0.8    | 9.14 [1.83, 38.26] | 0.0088 [0.0017, 0.0313] | 90.83 [61.72, 98.15] | 0.0040 [0.0007, 0.0231] | 99.977 [99.930, 99.991] | 0.0163 [0.0044, 0.0651] |
| 1.13   | 9.14 [1.83, 38.26] | 0.0088 [0.0017, 0.0313] | 90.83 [61.72, 98.15] | 0.0040 [0.0007, 0.0231] | 99.977 [99.930, 99.991] | 0.0163 [0.0044, 0.0651] |



(a) Focused log-log crossover.



(b) Full-range linear-scale growth.



(c) Direction-dependent odd squared-error share.

Figure 1: Signed-parity analysis of the vertical detector residual. The quadratic even and cubic odd nonlinear components cross near  $\rho = 1.31$ . For each direction, the bounded odd share is  $f = E_{\text{odd}}^2 / (E_{\text{odd}}^2 + E_{\text{even}}^2)$ ; the band reports its 10th, 50th, and 90th percentiles over the same 100 directions.

input set. The orbit-radius sweep then identifies a channel-specific failure of a purely quadratic residual model. Signed-direction measurements show that the vertical-only vertical residual is an odd cubic response overtaking a small even component, rather than a generic numerical loss of convergence. A direction-resolved four-sign reconstruction further separates the all-corrector quadratic source by orbit component: the horizontal orbit is dominated by the pure vertical block  $Q_{vv}$ , whereas the vertical orbit is dominated by the mixed block  $Q_{hv}$ . The remaining causal test is a sextupole/octupole attribution of the cubic coefficient.

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